INCH-POUND

MIL-DTL-27426/1
25 SEPTEMBER 1997

DETAIL SPECIFICATION

RINGS, RETAINING, SPIRAL
(UNIFORM CROSS SECTION)
MEDIUM DUTY EXTERNAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-27426, listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

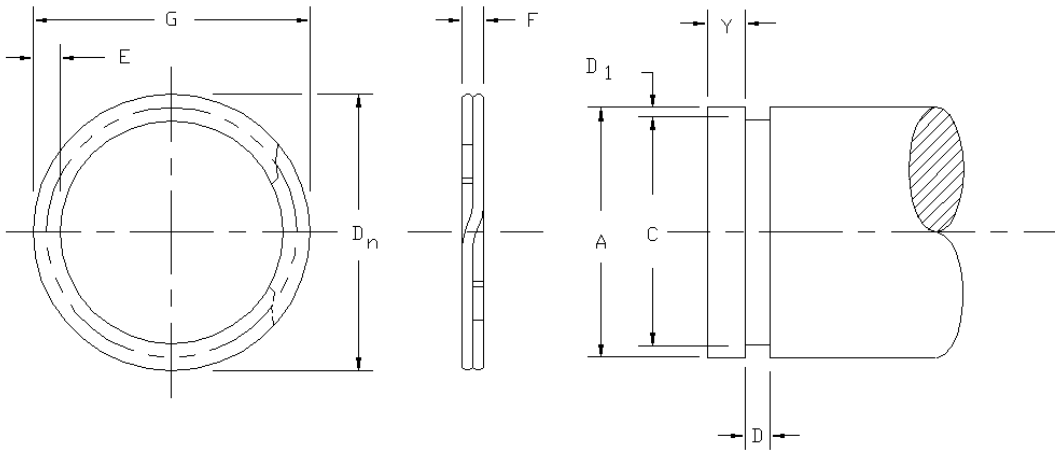


FIGURE 1. EXTERNAL RING

Maximum bottom groove radius is 0.005 inches for rings up to 0.946 inches free diameter and 0.010 inches for larger rings. Ring thickness applies only to unplated rings. Add 0.002 inches upper thickness tolerance for plated rings. Groove widths listed will accept rings without modification.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : the Defense Industrial Supply Center, (Code DISC-EED), 700 Robbins Avenue, Philadelphia, PA 19111-5096, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TABLE I. MEDIUM DUTY EXTERNAL RETAINING RINGS.

DASH NO.	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES A	MM	FREE DIAMETER G	TOL	THICKNESS F	TOL	RADIAL WALL E	TOL	DIAMETER C	TOL	WIDTH D	TOL	NOMINAL GROOVE DEPTH D_1
0100	.500	12.7	.467		.025		.045		.474		.030		.014
0101	.531	13.5	.498		.025		.045		.505		.030		.014
0102	.551	14.0	.518		.025		.045		.525	±.002	.030		.014
0103	.562	14.3	.529		.025		.045		.536		.030		.014
0104	.594	15.1	.561		.025		.045		.569		.030		.014
0105	.625	15.9	.585		.025		.055		.594		.030		.017
0106	.656	16.7	.617		.025		.055		.625		.030		.017
0107	.669	17.0	.629		.025		.055		.638		.030		.017
0108	.687	17.5	.647		.025		.055		.656		.030		.017
0109	.718	18.3	.679	+.000	.025		.055		.687		.030		.017
0110	.750	19.0	.710	−.013	.031		.065		.719		.036		.017
0111	.781	19.8	.741		.031		.065		.750		.036		.017
0112	.812	20.6	.771		.031		.065		.781		.036	+.003	.017
0113	.843	21.4	.803		.031		.065		.812	±.003	.036	−.000	.017
0114	.875	22.2	.828		.031		.065		.838		.036		.020
0115	.906	23.0	.860		.031		.065		.869		.036		.020
0116	.937	23.8	.889		.031		.065		.900		.036		.020
0117	.968	24.4	.916		.037		.075		.925		.042		.023
0118	.984	25.0	.930		.037		.075		.941		.042		.023
0119	1.000	25.4	.946		.037	±.002	.075		.957		.042		.023
0120	1.023	26.0	.968		.037		.075		.980		.042		.023
0121	1.031	26.3	.978		.037		.075		.988		.042		.023
0122	1.062	27.0	1.007		.037		.075	+.003	1.020		.042		.023
0123	1.093	27.8	1.040		.037		.075	−.005	1.051		.042		.023
0124	1.125	29.0	1.070		.037		.075		1.083		.042		.023
0125	1.156	39.3	1.102		.037		.075		1.114		.042		.023
0126	1.188	30.2	1.127		.043		.085		1.140		.048		.026
0127	1.218	31.0	1.159		.043		.085		1.170		.048		.026
0128	1.250	31.7	1.188		.043		.085		1.202	±.004	.048		.026
0129	1.281	32.6	1.221	+.000	.043		.085		1.233		.048		.026
0130	1.312	33.3	1.251	−.015	.043		.095		1.264		.048		.026
0131	1.343	34.1	1.282		.043		.095		1.295		.048		.026
0132	1.375	34.9	1.308		.043		.095		1.323		.048		.028
0133	1.406	35.8	1.340		.043		.095		1.354		.048		.028
0134	1.437	36.5	1.370		.043		.095		1.385		.048		.028
0135	1.468	37.3	1.402		.043		.095		1.416		.048		.028
0136	1.500	38.1	1.433		.043		.095		1.448		.048		.028
0137	1.562	39.6	1.490		.049		.108		1.507		.056	+.004	.030
0138	1.575	40.0	1.503		.049		.108		1.520		.056	−.000	.030
0139	1.625	41.7	1.549		.049		.108		1.566		.056		.032
0140	1.687	42.8	1.610		.049		.118		1.628		.056		.032
0141	1.750	44.4	1.673		.049		.118		1.691	±.005	.056		.032
0142	1.771	44.9	1.690		.049		.118		1.708		.056		.034
0143	1.813	46.0	1.730	+.000	.049		.118		1.749		.056		.034
0144	1.875	47.6	1.789	−.020	.049		.128		1.808		.056		.036
0145	1.938	49.2	1.844		.049		.128		1.861		.056		.036
0146	1.969	50.0	1.882		.049	+.003	.128		1.902		.056		.036
0147	2.000	50.8	1.909		.049		.128		1.929		.056		.038
0148	2.062	52.3	1.971		.049		.128		1.992		.056		.038
0149	2.125	53.9	2.029		.049		.128		2.051		.056		.040
0150	2.156	54.7	2.060		.049		.138		2.082		.056		.040
0151	2.165	55.0	2.070	+.000	.049		.138		2.091		.056		.040
0152	2.188	55.5	2.092	−.025	.049		.138		2.113	±.006	.056		.040
0153	2.250	57.1	2.153		.049		.138		2.176		.056		.040
0154	2.312	58.7	2.211		.049		.138		2.234		.056		.042

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TABLE I. MEDIUM DUTY EXTERNAL RETAINING RINGS. (CONT'D)

DASH NO.	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES A	MM	FREE DIAMETER G	TOL	THICKNESS F	TOL	RADIAL WALL E	TOL	DIAMETER C	TOL	WIDTH D	TOL	NOMINAL GROOVE DEPTH D_1
0155	2.362	59.9	2.261		.049		.138		2.284		.056		.042
0156	2.375	60.3	2.273		.049		.138		2.297		.056		.042
0157	2.437	61.9	2.331	+.000	.049		.148		2.355		.056		.044
0158	2.500	63.5	2.394	−.025	.049	+.003	.148		2.418		.056		.044
0159	2.559	64.9	2.449		.049		.148		2.473		.056	+.004	.046
0160	2.562	65.1	2.452		.049		.148		2.476		.056	−.000	.046
0161	2.625	66.6	2.514		.049		.148		2.539		.056		.046
0162	2.688	68.2	2.572		.049		.158		2.597		.056		.048
0163	2.750	69.8	2.635		.049		.158		2.660		.056		.048
0164	2.813	71.4	2.696		.049		.168		2.272		.056		.048
0165	2.875	73.0	2.755		.049		.168		2.781		.056		.050
0166	2.937	74.5	2.817		.049		.168		2.843		.056		.050
0167	2.952	74.9	2.831	+.000	.049		.168		2.858		.056		.050
0168	3.000	76.1	2.877	−.030	.061		.168		2.904		.068		.051
0169	3.062	77.7	2.938		.061		.168		2.966		.068		.051
0170	3.125	79.3	3.000		.061		.178		3.027		.068		.052
0171	3.149	79.9	3.023		.061		.178		3.051		.068		.052
0172	3.187	81.0	3.061		.061		.178		3.089		.068		.052
0173	3.250	82.5	3.121		.061		.178		3.150		.068		.053
0174	3.312	84.4	3.180		.061		.188		3.208		.068		.055
0175	3.343	84.9	3.210		.061		.188		3.239		.068		.055
0176	3.375	85.8	3.242		.061		.188		3.271		.068		.055
0177	3.437	87.2	3.301		.061		.188		3.331		.068		.056
0178	3.500	88.8	3.363		.061		.188		3.394		.068		.056
0179	3.543	89.9	3.402		.061		.198		3.433		.068		.058
0180	3.562	90.5	3.422		.061		.198		3.452	±.006	.068		.058
0181	3.625	92.0	3.483		.061		.198		3.515		.068		.058
0182	3.687	93.6	3.543		.061		.198		3.575		.068		.059
0183	3.740	95.0	3.597		.061		.198		3.628		.068		.059
0184	3.750	95.2	3.606		.061		.198		3.638		.068		.059
0185	3.812	97.1	3.668		.061		.198	+.004	3.700		.068		.059
0186	3.875	98.3	3.724	+.000	.061		.208	−.006	3.757		.068		.062
0187	3.938	99.9	3.784	−.040	.061		.208		3.820		.068		.062
0188	4.000	101.5	3.842		.061		.218		3.876		.068		.065
0189	4.063	103.2	3.906		.061		.218		3.939		.068		.065
0190	4.125	104.8	3.967		.061		.218		4.000		.068	+.005	.065
0191	4.134	105.0	3.975		.061		.218		4.010		.068	−.000	.065
0192	4.188	106.5	4.030		.061		.218		4.058		.068		.065
0193	4.250	107.9	4.084		.061		.228		4.120		.068		.068
0194	4.312	109.5	4.147		.061		.228		4.182		.068		.068
0195	4.331	110.0	4.164		.061		.228		4.200		.068		.068
0196	4.375	111.0	4.208		.061		.228		4.245		.068		.068
0197	4.437	112.7	4.271		.061		.228		4.307		.068		.068
0198	4.500	114.2	4.326		.061		.238		4.364		.068		.071
0199	4.562	116.0	4.384		.072		.250		4.442		.079		.073
0200	4.625	117.4	4.447		.072		.250		4.485		.079		.073
0201	4.687	119.0	4.508		.072		.250		4.547		.079		.073
0202	4.724	120.0	4.546		.072		.250		4.584		.079		.073
0203	4.750	120.6	4.571		.072		.250		4.610		.079		.073
0204	4.812	122.2	4.633		.072	±.004	.250		4.672		.079		.073
0205	4.875	123.8	4.695		.072		.250		4.735		.079		.073
0206	4.937	125.4	4.757		.072		.250		4.797		.079		.073
0207	5.000	126.9	4.820		.072		.250		4.856		.079		.075
0208	5.118	130.0	4.934		.072		.250		4.974		.079		.075
0209	5.125	130.1	4.939		.072		.250		4.981		.079		.075
0210	5.250	133.2	5.064	+.000	.072		.250		5.107		.079		.075
0211	5.375	136.5	5.187	−.050	.072		.250		5.228	±.007	.079		.077
0212	5.500	139.9	5.308		.072		.250		5.353		.079		.077
0213	5.511	140.0	5.320		.072		.250		5.364		.079		.077

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DASH NO.	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES A	MM	FREE DIAMETER G	TOL	THICKNESS F	TOL	RADIAL WALL E	TOL	DIAMETER C	TOL	WIDTH D	TOL	NOMINAL GROOVE DEPTH D_1
0214	5.625	142.8	5.433		.072		.250	+.004	5.478	±.007	.079	+.005	.077
0215	5.750	145.9	5.550	+.000	.072		.250	−.006	5.597		.079	−.000	.080
0216	5.875	149.2	5.674	−.050	.072		.250		5.722		.079		.080
0217	5.905	150.0	5.705		.072		.250		5.752		.079		.080
0218	6.000	152.3	5.798		.072		.250		5.847		.079		.080
0219	6.125	155.6	5.903		.086		.312		5.953		.094		.090
0220	6.250	158.8	6.026		.086		.312		6.078		.094		.090
0221	6.299	160.0	6.076		.086		.312		6.127		.094		.090
0222	6.375	162.0	6.152		.086		.312		6.203		.094		.090
0223	6.500	165.1	6.274		.086	±.004	.312		6.328		.094		.090
0224	6.625	168.3	6.390	+.000	.086		.312		6.443		.094		.095
0225	6.750	171.4	6.513	−.060	.086		.312		6.568		.094		.095
0226	6.875	174.6	6.638		.086		.312		6.693		.094		.095
0227	7.000	177.8	6.761		.086		.312		6.818		.094		.095
0228	7.125	181.0	6.877		.086		.312	+.004	6.933		.094		.100
0229	7.250	184.1	6.999		.086		.312	−.008	7.058		.094		.100
0230	7.375	187.3	7.125		.086		.312		7.183		.094		.100
0231	7.500	190.5	7.250		.086		.312		7.308	±.008	.094		.100
0232	7.625	193.7	7.363		.086		.312		7.423		.094	+.006	.105
0233	7.750	196.8	7.486		.086		.312		7.548		.094	−.000	.105
0234	7.875	200.0	7.611		.086		.312		7.673		.094		.105
0235	8.000	203.0	7.734		.086		.312		7.798		.094		.105
0236	8.250	209.5	7.972		.086		.375		8.038		.094		.110
0237	8.500	216.0	8.220		.086		.375		8.288		.094		.110
0238	8.750	222.2	8.459	+.000	.086		.375		8.528		.094		.115
0239	9.000	228.6	8.707	−.070	.086		.375		8.778		.094		.115
0240	9.250	235.0	8.945		.086		.375		9.018		.094		.120
0241	9.500	241.0	9.194		.086		.375		9.268		.094		.120
0242	9.750	247.8	9.432		.086		.375		9.508		.094		.125
0243	10.000	254.0	9.680		.086		.375		9.758		.094		.125
0244	10.250	260.7	9.918		.086		.375		9.998		.094		.130
0245	10.500	265.0	10.166		.086		.375		10.248		.094		.130
0246	10.750	273.0	10.405		.086		.375		10.488		.094		.135
0247	11.000	279.0	10.653		.086		.375		10.738		.094		.135

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